

// Day 49 of 60

API DESIGN

gRPC and Protocol Buffers

→ Swipe through all 6 slides



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REST Is Slow for Microservices

Internal traffic deserves better

REST issues at scale:

- JSON parsing is slow
- Repeated text headers
- HTTP/1.1 head-of-line
- No native streaming

```
// Fine for browsers,  
// painful for service-to-  
// service chatter.
```



JSON over the wire is wasteful



Enter gRPC

Built by Google for internal RPC

gRPC stack:

- HTTP/2 (multiplexed)
- Protocol Buffers (binary)
- Generated stubs
- Bi-directional streaming

Result:

7-10x faster than REST
smaller payloads
type safety



Designed for backend-to-backend



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What Are Protobuf?

Binary, compact, fast

JSON:

```
{"name": "John", "age": 30}
```

→ 24 bytes of text

Protobuf:

```
0a 04 4a 6f 68 6e 10 1e
```

→ 8 bytes binary

// 3x smaller

// 10x faster to parse



Binary formats win at scale



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The .proto Contract

One file, every language

```
// user.proto
service UserService {
  rpc GetUser(UserId)
    returns (User);
}

message User {
  string name = 1;
  int32 age = 2;
}

// Generates client + server
// in Java, Go, Python, ...
```



Schema-first APIs catch bugs early



Why It's Fast

Four wins, stacked

1. HTTP/2 multiplexing
→ many calls, one conn
2. Binary serialization
→ smaller payloads
3. Schema-driven parsing
→ no field name lookups
4. Code-gen stubs
→ no reflection cost



Each layer compounds the speedup



When NOT to Use It

gRPC isn't for browsers

Avoid gRPC for:

- Public web APIs
- Browser-to-server
- Simple integrations

Reason:

Browsers don't natively
speak HTTP/2 + Protobuf
(need gRPC-Web proxy)

```
// Internal services? Yes.  
// Public website? No.
```



Right tool for right boundary



done = true

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